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Studierichting: Informatica
Oefeningenreeks 1D nr. 19.12

1. Oefening:
$$z^5 + 2z^4 + 2z^3 + 4z^2 - 15z - 30$$
2. Ontbinding in factoren:
$$z^4(z+2) + 2z^2(z+2) - 15(z+2) = 0$$
$$(z^4 + 2z^2 - 15)(z+2) = 0$$
   - $z + 2 = 0 \Rightarrow z = -2$
   - $z^4 + 2z^2 - 15 = 0$
3. Substitueer $w = z^2$ in $z^4 + 2z^2 - 15 = 0$:
$$w^2 + 2w - 15 = 0$$
$$(w+5)(w-3) = 0$$
   Dit geeft:
   - $w + 5 = 0 \Rightarrow w = -5 \Rightarrow z^2 = -5 \Rightarrow z = \pm i\sqrt{5}$
   - $w - 3 = 0 \Rightarrow w = 3 \Rightarrow z^2 = 3 \Rightarrow z = \pm\sqrt{3}$
4. Nulpunten in $\mathbb{C}$:
$$z = -2, \quad z = i\sqrt{5}, \quad z = -i\sqrt{5}, \quad z = \sqrt{3}, \quad z = -\sqrt{3}$$
5. Eindresultaat $\mathbb{R}[z]$:
$$(z+2)(z^2+5)(z+\sqrt{3})(z-\sqrt{3})$$

References